SECURITY BOT: AUTONOMOUS PERSON TRACKING AND PATROL MONITORING

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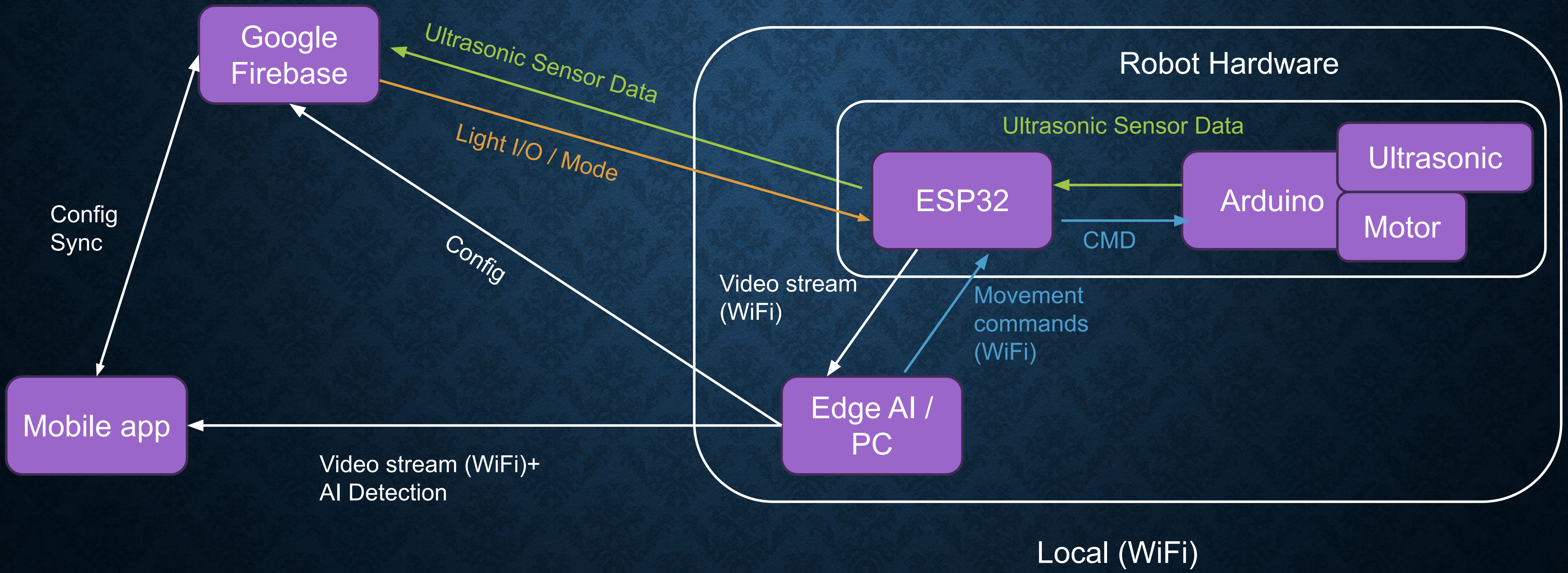
BRIEF DESCRIPTION

- This project presents an autonomous AlphaBot for security monitoring. An ESP32-CAM streams video to a PC, where a YOLO-based pipeline detects and tracks a person in real time. The robot supports four app-controlled modes: IDLE, PATROL ONLY, FOLLOW ONLY, and PATROL + FOLLOW, which combines patrol with automatic follow behavior after detection. An ultrasonic sensor improves obstacle awareness, and an Android app provides telemetry, mode switching, and remote light control.

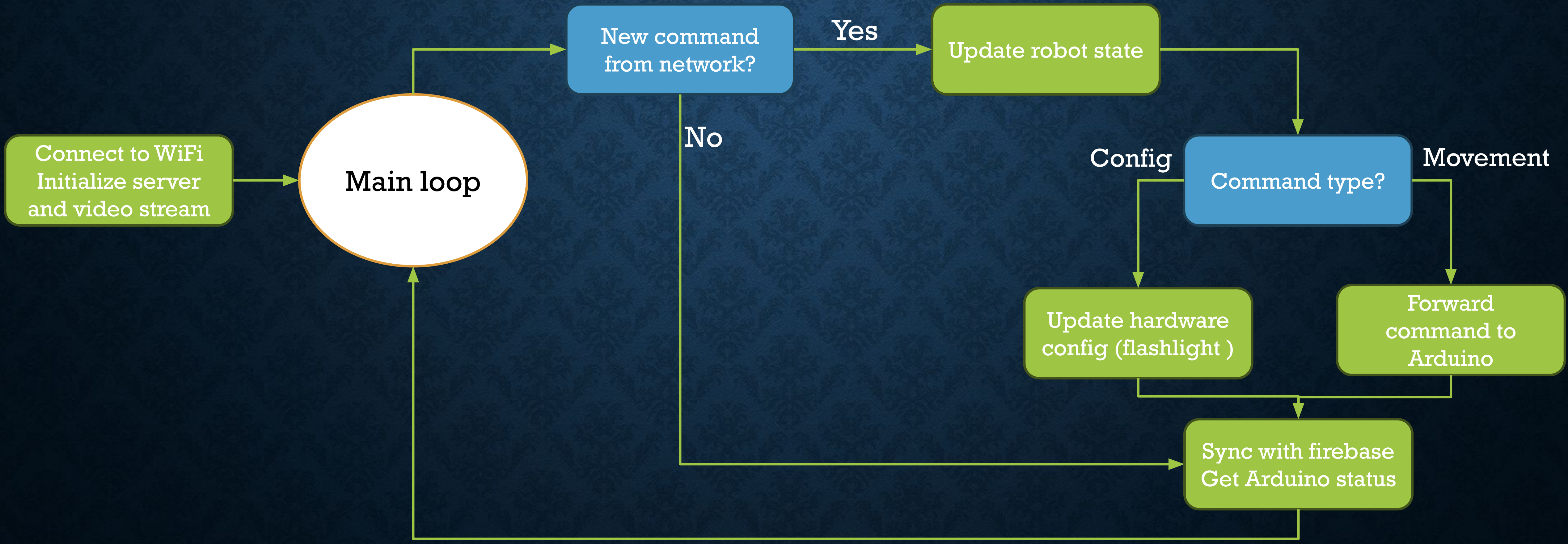
SIMILAR COMMERCIAL PRODUCTS

- Knightscope
  - o K5 Autonomous Security Robot
- Kabam Robotics
  - o HALO
    Outdoor Autonomous Mobile Robot
- Many more

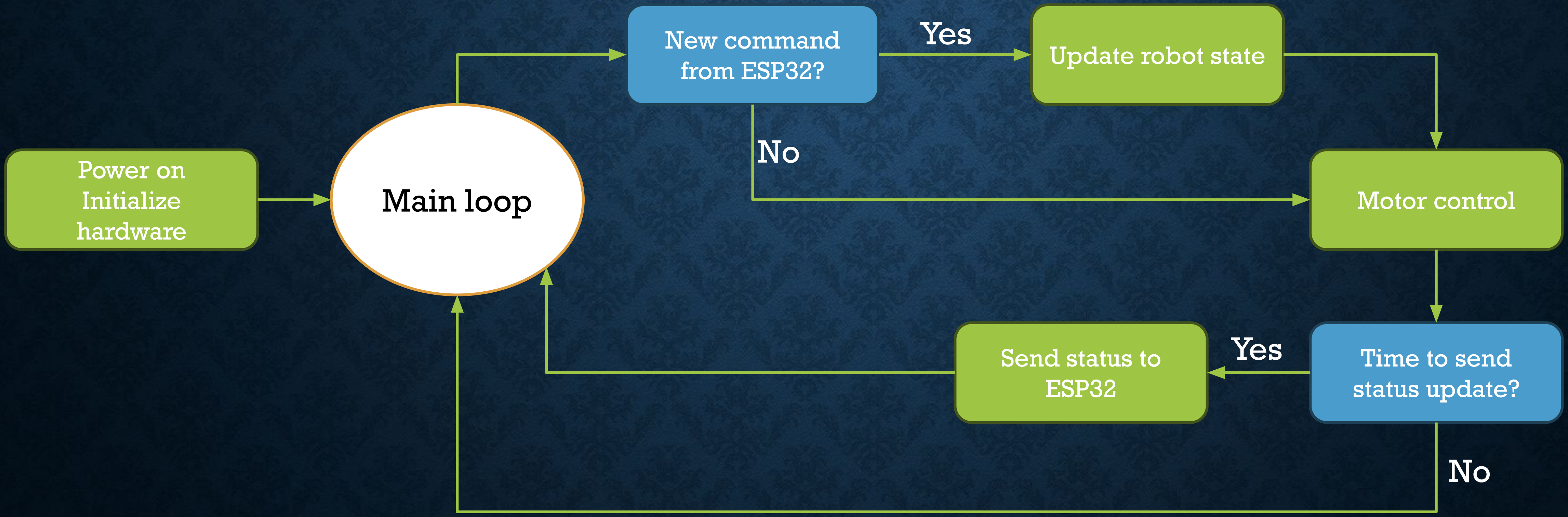
BLOCK DIAGRAM



MCU FLOW DIAGRAM – ESP32

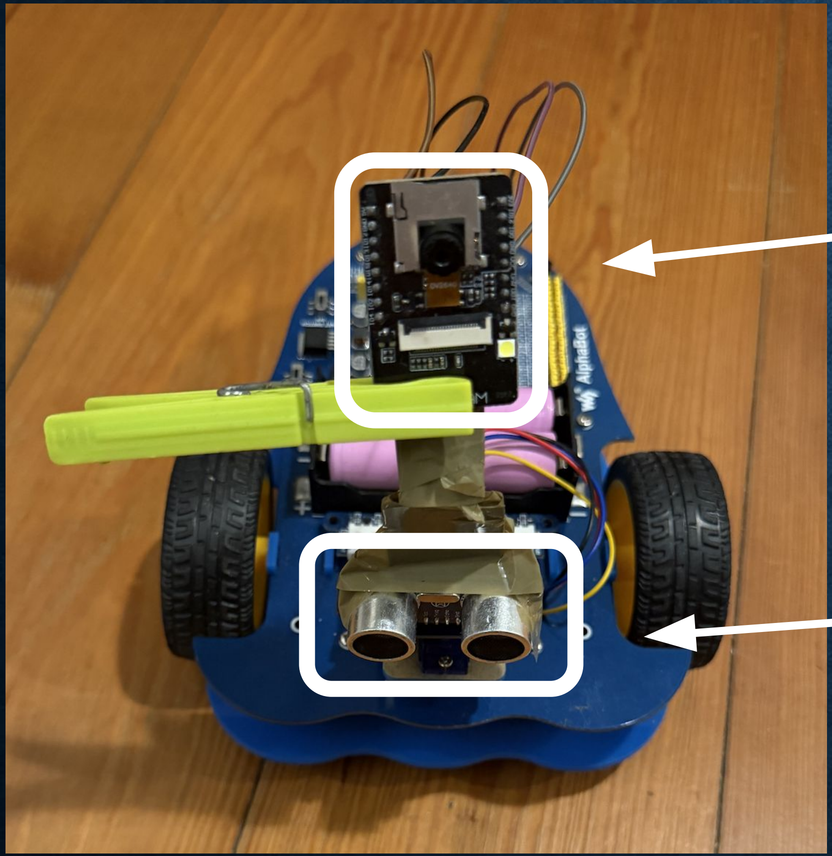


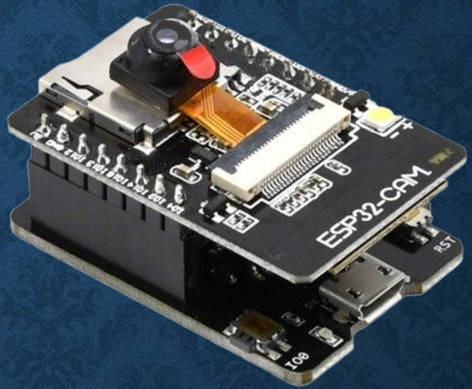
MCU FLOW DIAGRAM – ARDUINO



MACHINE LEARNING

PHOTO OF HARDWARE





ESP 32 CAM

Ultrasonic Sensor

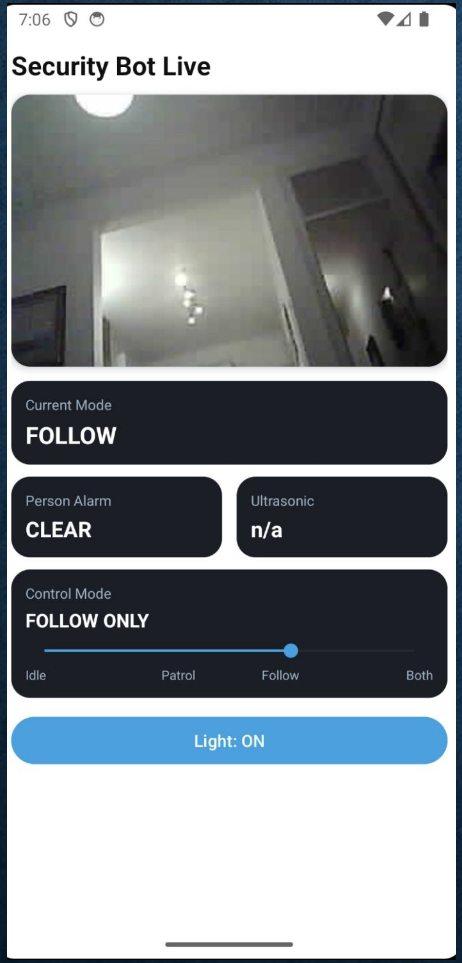
BRIEF DESCRIPTION

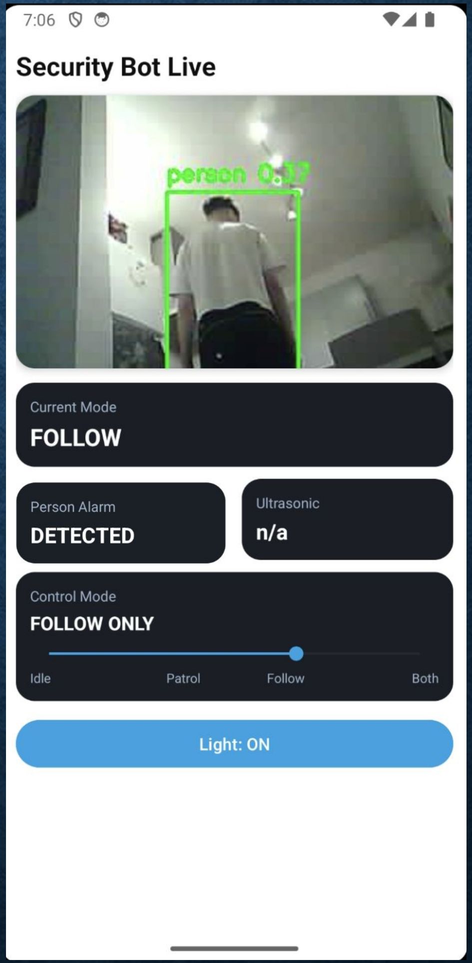
- mode_profile
  - IDLE, PATROL_ONLY, FOLLOW_ONLY, PATROL_FOLLOW
- mode:
  - INIT, IDLE, PATROL, FOLLOW, HOLD FOR x.xS
- Tracking_alarm:
  - true = DETECTED, false = CLEAR
- cmd:
  - Format: M + direction + speed
- ultrasonic_cm:
  - number in cm (for example 18.4)
- tracking_state::
  - remote: idle, obstacle 0.0cm -> avoid, tracking:centered -> forward | seek=False | cmd=MF70, offline

MOBILE APP FUNCTIONALITY

- View camera stream
- Enable/disable flashlight
- Set and view operating mode (patrol, following)
- View alarm status
- View ultrasonic sensor distance

MOBILE APP SNAPSHOTS





CONCLUSION AND IMPROVEMENTS

- To be written…